AURA TECHNICAL FEST
ALIAH UNIVERSITY

SMART WALKING STICK

Detect Hazards • Detect Water • Stay Safe

ECE
INNOVATION FOR A SAFER TOMORROW

THE PROBLEM

Visually impaired people face multiple hazards while walking — obstacles in front, uneven surfaces like steps or potholes, and water or puddles on the path. These hazards can lead to falls, injuries and loss of independence.

They need a smarter, safer and more independent way to move.

OUR SOLUTION

SMARTGUARD is a smart walking stick that detects front obstacles, downward hazards (steps/potholes) and water/puddles, and provides real-time vibration and audio alerts to help visually impaired users navigate safely and independently.

KEY INNOVATION

Hazard Detection
Detects obstacles, steps, & potholes.

Water Detection
Detects puddles or water on the path.

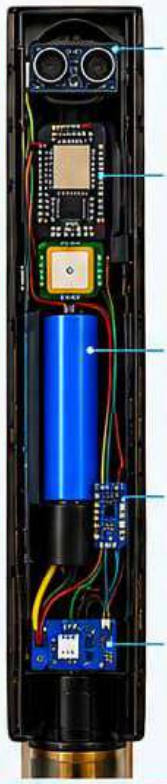
Simple Alerts
Vibration + audio alerts for instant response.

EXTERNAL VIEW



- Handle & Grip (with wrist strap)
- SOS Button (emergency alert)
- Vibration Motor 1 (front hazard alert)
- Vibration Motor 2 (water alert)
- Main Control Unit (ESP32 inside)
- Ultrasonic Sensor (detects front obstacles)
- Water Sensor (detects water/puddles)
- Rubber Tip (anti-slip)

INSIDE VIEW



- Ultrasonic Sensor (front detection)
- ESP32 (main controller)
- Battery (Li-ion 18650)
- Water Sensor (module)
- Wiring & Power Management

KEY FEATURES

Obstacle Detection (Front Ultrasonic/ToF)
Detects objects ahead and alerts the user.

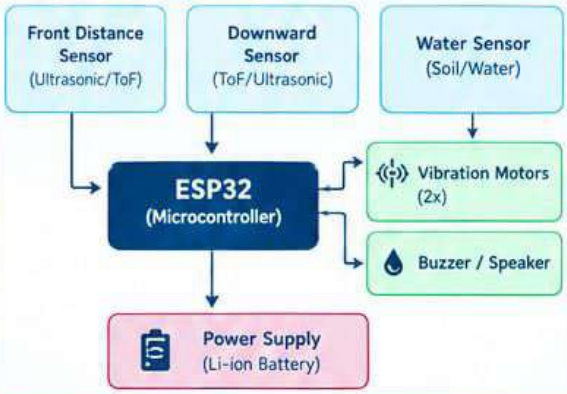
Step / Pothole Detection (Downward Sensor)
Detects uneven ground and steps, alerts the user.

Water Detection (Water Sensor)
Detects puddles or water on the path, alerts the user.

Alert System
Vibration + Audio
Different patterns for each hazard.

Safer Path. Independent Steps.

SYSTEM BLOCK DIAGRAM



PROTOTYPE (3D RENDER)



WORKING PRINCIPLE

1. Front sensor detects obstacles (ultrasonic/ToF). → Vibration + Audio Alert (for front hazard)
2. Downward sensor detects steps or potholes (ToF/Ultrasonic). → Vibration + Audio Alert (for downward hazard)
3. Water sensor detects puddles or water on the path. → Vibration + Audio Alert (for water hazard)

DEMONSTRATION PLAN

1. Show front obstacle detection (block/board/object).
2. Show step/pothole detection (step/raised platform).
3. Show water detection (water puddle on floor).
4. Real-time alerts: vibration + audio.


Obstacle Detection


Step / Pothole Detection


Water Detection

FUTURE ENHANCEMENTS

We aim to make SMARTGUARD smarter, safer and more useful in the future.

- **GPS Location Tracking** (helps in emergency and family tracking)
- **Mobile App Integration** (settings, alerts, location, battery status)
- **AI-Based Hazard Recognition** (better accuracy with machine learning)
- **Solar Charging** (extended battery life)
- **Multi-Language Alerts** (voice + text in local languages)
- **Foldable & Lightweight Design** (easy to carry and store)

A small step towards a safer and independent future ♥